Intelligent Data Analytics

Infectious Disease Modelling

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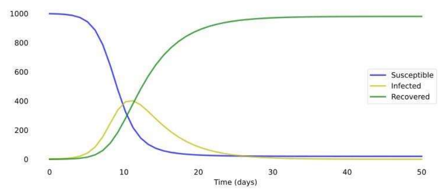
Infectious Disease Modelling - SIR



SIR model

- **N:** total population
- **S(t):** number of people susceptible on day t
- **I(t):** number of people infected on day t
- **R(t):** number of people recovered on day t
- **β:** expected amount of people an infected person infects per day
- **D:** number of days an infected person has and can spread the disease
- **γ:** the proportion of infected recovering per day (γ = 1/D)
- **R_0:** the total number of people an infected person infects ($R_0 = \beta / \gamma$)

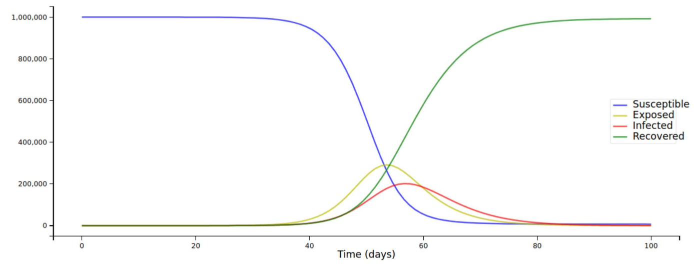
SIR model

- N = 100
- β=1, D=7 and γ=1/7
- Let's say that on day t, 60 people are infected (so I(t)=60), and 30 people are still susceptible (so S(t)=30 and R(t)=100–60–30=10).
- Now, how do S(t) and I(t) and R(t) change to the next day?
  - Change of S(t) to the next day S'(t) = - β · I(t) · S(t) / N.
  - I'(t) = β · I(t) · S(t) / N -γ · I(t)
  - R'(t) = γ · I(t)

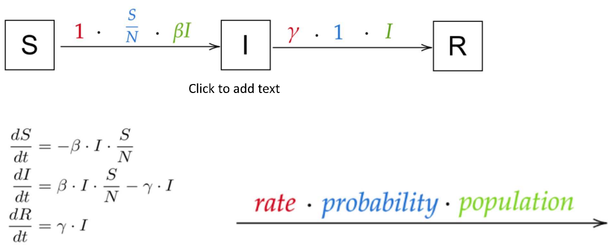
SIR model - *ordinary differential equations*

$$\frac{dS}{dt} = -\beta \cdot I \cdot \frac{S}{N}$$
$$\frac{dI}{dt} = \beta \cdot I \cdot \frac{S}{N} - \gamma \cdot I$$
$$\frac{dR}{dt} = \gamma \cdot I$$

Beyond the basic SIR



Models as State Transitions



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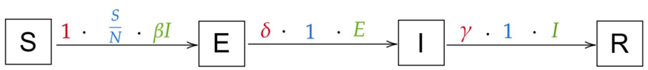
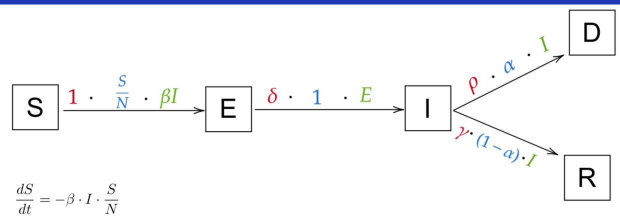
$$\frac{dS}{dt} = -\beta \cdot I \cdot \frac{S}{N}$$
$$\frac{dI}{dt} = \beta \cdot I \cdot \frac{S}{N} - \gamma \cdot I$$
$$\frac{dR}{dt} = \gamma \cdot I$$

SEIR



$$\frac{dS}{dt} = -\beta \cdot I \cdot \frac{S}{N}$$
$$\frac{dE}{dt} = \beta \cdot I \cdot \frac{S}{N} - \delta \cdot E$$
$$\frac{dI}{dt} = \delta \cdot E - \gamma \cdot I$$
$$\frac{dR}{dt} = \gamma \cdot I$$

Deriving the Dead-Compartment



$$\frac{dS}{dt} = -\beta \cdot I \cdot \frac{S}{N}$$
$$\frac{dE}{dt} = \beta \cdot I \cdot \frac{S}{N} - \delta \cdot E$$
$$\frac{dI}{dt} = \delta \cdot E - (1-\alpha) \cdot \gamma \cdot I - \alpha \cdot \rho \cdot I$$
$$\frac{dR}{dt} = (1-\alpha) \cdot \gamma \cdot I$$
$$\frac{dD}{dt} = \alpha \cdot \rho \cdot I$$

Time-Dependent Variables

- On day *L*, a strict "lockdown" is enforced => pushing R_0 to 0.9

```
def R_0(t):
    return 5.0 if t < L else 0.9
```

```
def beta(t):
    return R_0(t) * gamma
```


Colab

- https://colab.research.google.com/drive/16uL8ASmEnBxUqmQX4-aSfNPlR0Rp8-Eh?usp=sharing

Thanks for your attention!